TOFUNMI OGUNFUNMI

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EDUCATION

Ph.D. University of California, Berkeley Expected December 2026
Materials Science & Engineering •Advisor: Dr. Mary Scott • GEM Fellow • GPA: 3.64
GEM National Consortium Full Fellowship • McDermott Scholar
Areas of Expertise: Solid-state battery materials, electron microscopy, materials analysis & characterization

M.S. Worcester Polytechnic Institute September 2020
Materials Science & Engineering

B.S. Massachusetts Institute of Technology May 2018
Materials Science & Engineering

TECHNICAL SKILLS

Materials Development & Synthesis: Solid-state electrolyte synthesis, Air-sensitive sample preparation

Characterization: SEM/EDS, TEM/STEM, EELS, cryo-FIB, FIB-SEM tomography, XRD, XAS

Data Analysis & Statistical Methods: Python; Machine Learning; Data visualization techniques SQL, Tableau, Figma; DoE principles, statistical data analysis

Battery Testing: Impedance spectroscopy, cyclic voltammetry, electrochemical performance–structure correlation

Communication & Documentation: Peer-reviewed publications, lab protocol development, technical reporting, cross-functional collaboration

PROFESSIONAL EXPERIENCE

Lawrence Berkeley National Laboratory (LBNL) Aug 2021 – Present
Graduate Researcher, National Center for Electron Microscopy | Advisor: Dr. Mary Scott

Materials Development: Ag/Li_3PS_4 Solid-State Electrolytes

- Led multimodal characterization (SEM/EDS, FIB tomography) of Ag-nanoparticle-filled Li_3PS_4 electrolytes to establish structure–property relationships driving dendrite suppression.
- Demonstrated that dispersing Ag nanoparticles into amorphous Li_3PS_4 solid electrolyte suppresses lithium dendrite failure, extending cell lifetime from ~18 h to 80 h and enabling stable cycling.

Data-Driven Process Development: 4D-STEM Analysis Pipeline

- Built Python workflows (py4DSTEM, NumPy, scikit-learn) to process GB-scale 4D-STEM datasets of LLZO, applying virtual diffraction imaging and PCA-based clustering to resolve grain microstructure
- Developing 4D-STEM strain mapping of LLZO to quantify local lattice strain linked to mechanical failure and Li-dendrite propagation

Beam–Material Interactions & High-Throughput Test Protocols

- Investigated electron-beam–induced damage on LLZO electrolyte using air-free transfer; contributed to high-throughput conductivity-screening protocols for accelerated materials qualification.
- Performed ex-situ characterization (TEM/SEM) of LLZO to support work showing that electrostatic beam charging drives lithium growth on solid-electrolyte surfaces, establishing criteria to separate genuine electrochemistry from imaging artifacts;

IBM, Chemical & X-ray Failure Analysis Summer 2018 & Summer 2019
Hardware Developer Intern • Fishkill, NY

- Performed root-cause failure analysis on field-returned Li-ion battery leakage by 3D X-ray and SEM/EDS, localizing the leak to the gasket and identifying corrosion- and electrolyte-derived chemistry.
- Calibrated a high-temperature XRD stage against a silicon thermal-expansion standard (25–275 °C) to enable accurate HTXRD phase analysis of flip-chip solder
- Modeled SnO_2-on-silicon films by X-ray reflectivity (XRR), resolving layer thickness, density, and roughness and flagging an unexpected dense surface layer for XRF follow-up

Worcester Polytechnic Institute, Soboyejo Energy Lab Aug 2018 – Sep 2020
Graduate Research Assistant | Advisor: Dr. Wole Soboyejo

- Synthesized and characterized a novel $LiBH_4$–$LiNH_2$ -based solid electrolyte (XRD, SEM/EDS, impedance, CV), correlating LiI doping with amorphization and ionic conductivity
- Mapped compressive failure via in-situ DIC, linking LiI content to a brittle-to-ductile transition in failure mode and pinpointing an optimal crimping pressure for cell performance

MIT Photovoltaics Research Lab Feb 2017 – Jun 2018

Undergraduate Researcher | Advisors: Dr. Juan-Pablo Correa-Baena, Dr. Tonio Buonassisi

- High-throughput XRD characterization of 75 perovskite-inspired thin-film compositions in an ML-accelerated discovery workflow

SELECTED PUBLICATIONS & PRESENTATIONS

Ogunfunmi, T., Diallo, M.S., et al. (2025). Mitigating Battery Cell Failure: Role of Ag-Nanoparticle Fillers in Solid Electrolyte Dendrite Suppression. *Advanced Energy Materials*. Co-first author.

Avvaru, V.S., **Ogunfunmi, T.**, et al. (2025). Tin–Carbon Dual Buffer Layer to Suppress Lithium Dendrite Growth in All-Solid-State Batteries. *ACS Nano* 19(18), 17347–17356.

Ko, Y., Baird, M.A., Peng, X., **Ogunfunmi, T.**, et al. (2024). Omics-enabled understanding of electric aircraft battery electrolytes. *Joule* 8(8), 2393–2411.

Cheng, J., Peng, X., Zhang, Y.Q., Tian, Y., **Ogunfunmi, T.**, et al. (2024). Oxygen transport through amorphous cathode coatings in solid-state batteries. *Chemistry of Materials* 36(6), 2642–2651.

Byeon, Y.W., Yang, S., **Ogunfunmi, T.**, et al. (2024). Conductive carbon embedded beneath cathode active material for longevity of solid-state batteries. *Journal of Materials Chemistry A* 12(14), 8359–8369.

Peng, X., Tu, Q., **Ogunfunmi, T.**, et al. (2023). Unraveling Li growth kinetics in solid electrolytes due to electron beam charging. *Science Advances* 9(17), eabq3285.

Ogunfunmi, T., et al. (2021). An investigation into compressive deformation and failure mechanisms in a novel Li-ion solid-state electrolyte. *MRS Advances* 6(4), 154–161.

Sun, S., … **Ogunfunmi, T.**, et al. (2019). Accelerated development of perovskite-inspired materials via high-throughput synthesis and machine-learning diagnosis. *Joule* 3(6), 1437–1451.

Conference presentations: MRS Fall 2020 (oral); M&M 2023, 2024 (poster).

AWARDS & AFFILIATIONS

- Gatan EELS School Graduate
- MSA Educational Outreach Team & Regional Liaison
- BGESS Financial Officer
- MRS Mentor & Mentee
- NSBE Bay Bridge Mentee